Federal State Autonomous Educational Institution
for the Higher Education
National Research University "Higher School of Economics"
Faculty of Mathematics

Stanislav Demurov

Uniqueness theorems for non-compact electrostatic manifolds with boundary

Master's thesis

Field of study: *01.04.01 "Mathematics"*
Master's programme "Mathematics"

Reviewer:
Doctor of Sciences, professor
Alexei Penskoi

Scientific supervisor:
PhD, associate professor
Vladimir Medvedev

Moscow 2025

Abstract

Static manifolds with boundary were recently introduced by Cruz and Vitório in the context of the prescribed scalar curvature problem in a manifold with boundary with prescribed mean curvature [CV19]. These manifolds are also interesting from the point of view of the general theory of relativity, where they arise as static spacetimes satisfying Einstein equations.

In this thesis, we introduce electrostatic manifolds with boundary as a natural generalization of the static manifolds with boundary in the presence of a nontrivial electric field. We study their general properties and prove uniqueness theorems for the non-compact case, relating electrostatic manifolds with boundary to Reissner–Nordström manifolds.

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1 Introduction

An *electrostatic manifold with boundary* is a quadruple (M^n, g, V, E), where (M^n, g) is a Riemannian manifold with boundary, which admits a nontrivial smooth solution $V : M \to \mathbb{R}$, called the *static potential*, to the following system

$$\begin{cases} \operatorname{Hess}_g V - (\Delta_g V)\, g - V \operatorname{Ric}_g = 2V \left(\mathrm{E}^\flat \otimes \mathrm{E}^\flat - |\mathrm{E}|_g^2\, g \right) & \text{in } M, \\ \dfrac{\partial V}{\partial \nu} g - V B_g = 0 & \text{on } \partial M, \end{cases}$$

where $E \in \Gamma(TM)$ is a tangent vector field called *electric field*, $E^\flat = g(E, \cdot) \in \Gamma(T^*M)$ is the one-form, metrically dual to E, B_g is the second fundamental form of ∂M with respect to the outward unit normal vector field ν.

By setting the electric field E equal to zero, the aforementioned system of equation reduces to the case of static manifolds with boundary, which were recently introduced by Cruz and Vitório in [CV19] in the context of finding a Riemannian metric with prescribed curvature. Also, from the perspective of the general theory of relativity electrostatic manifolds with boundary arise as a solution of the Einstein–Maxwell equations which are the generalization of the Einstein equations. Therefore, electrostatic manifolds with boundary are a natural generalization of the static manifolds with boundary and so they must also generalize their properties.

For example as in the static case, a metric on an electrostatic manifold with boundary (M^n, g_o, V, E) is a critical point of the following functional

$$\mathcal{F}[g] = \int_M V \left(R_g - 6|E|_g^2 \right) dv_{g_o} + 4 \int_M V |E|_g^2\, dv_g + 2 \int_{\partial M} V H_g\, ds_{g_o}$$

on the set of Riemannian metrics on M, which we denote by $\mathcal{R}(M)$ (see Theorem 2.6 below).

Also, the boundary of an electrostatic manifold with boundary is totally umbilical, and the zero-level set of the potential is an embedded totally geodesic hypersurface, and, due to these properties, it is natural to relate them to objects coming from physics such as photon surfaces and horizons of black holes (respectively).

A canonical example of an electrostatic manifold with boundary is the *Reissner-Nordström manifold with boundary* (see Appendix 5 for the derivation of the photon sphere equations as solutions to the boundary condition). We thoroughly describe these manifolds in Section 3.1.

If (M^n, g, V, E) is an asymptotically flat electrostatic manifold (Section 3.2), then it is natural to assume that there exists a smooth function Ψ on M^n such that $VE = -\nabla^g \Psi$ (see Section 3). Under this assumption and by analogy with the static case (see Theorem 1.18 in [Med24]), we prove the following uniqueness theorems

Theorem 1.1. Let (M^n, g, V, E) be a complete (up to the boundary) one-ended asymptotically Reissner–Nordström system of mass m and charge q (see Section 3.2), which is an electrostatic manifold with compact boundary. Then

I. If all the connected components of ∂M are subextremal (i.e. satisfying (3.3)), $V \neq 0$ on $M \cup \partial M$, $H_g < 0$, $V = const.$ on ∂M, $E \in \Gamma(\mathrm{N}\partial M)$, and $|\mathrm{E}| = const.$ on ∂M, then (M, g) is isometric to RN^n_- with $m > |q|$, $V = V_{m,q}$, $E = E_{m,q}$.

II. If ∂M is connected and subextremal, $n = 3$, $\Sigma = V^{-1}(0) \subset \operatorname{Int} M$ is compact and separates the boundary and the end, $\nu(V) = const. \neq 0$ on Σ, $V = const. \neq 0$ on ∂M, $E \in \Gamma(\mathrm{N}\partial M)$, $|\mathrm{E}| = const.$ on ∂M, and $H_g > 0$, then (M, g) is isometric to RN^3_+ with $m > |q|$, $V = V_{m,q}$ and $E = E_{m,q}$.

Here $V_{m,q}$ is the Reissner–Nordström static potential in the isotropic coordinates

$$V_{m,q}(s) = \frac{\left(1 - \frac{m^2-q^2}{4s^{2(n-2)}}\right)}{\left(1 + \frac{m+q}{2s^{n-2}}\right)\left(1 + \frac{m-q}{2s^{n-2}}\right)}$$

and $E_{m,q}$ is the Reissner–Nordström electric field in the isotropic coordinates

$$E_{m,q}(s) = \frac{n-2}{\mathfrak{C}_n} \frac{q\left(1 - \frac{m^2-q^2}{4s^{2(n-2)}}\right)}{s^{n-1}\varphi^{2n-3}_{m,q}(s)\left(\varphi_{m,q}(s) + s\varphi'_{m,q}(s)\right)}\partial_s.$$

Also, using theory from [BCC24], we obtain the following result for the 3-dimensional case with weaker assumptions on asymptotic decay

Theorem 1.2. Let (M^3, g, V, E) be a complete (up to the boundary) one-ended asymptotically flat electrostatic system (see Section 3.2), which is an electrostatic manifold with compact connected boundary. Then, assume that on ∂M the inequality $V^2 \geqslant |1 - \Psi^2|$ holds if $V^2 > (1 - |\Psi|)^2$ and that $V = 1$ and $\Psi = 0$ do not both hold. If $V \neq 0$ on $M \cup \partial M$, $H_g < 0$, $V = const.$ on ∂M, $E \in \Gamma(N\partial M)$, and $|E| = const.$ on ∂M, then (M, g) is isometric to RN^3_-, $V = V_{m,q}$, $E = E_{m,q}$ with $m > |q|$ or $m = |q| \neq 0$ or $m < |q|$.

Remark 1.1. All manifolds in this paper are assumed to be orientable and of dimension $n \geqslant 3$.

Remark 1.2. We will use ∇^g to denote the gradient operator with respect to the metric g.

1.1 Thesis organization

The thesis is organized as follows. In Section 2.1 we derive the defining equations of electrostatic manifolds from the source-free Einstein–Maxwell equations in static spacetimes. In Section 2.2 we show how the defining equations for electrostatic manifolds with boundary could be rewritten in the form of second order overdetermined elliptic equation, then we show how that metrics on electrostatic manifolds with boundary arise as critical points of an appropriate variational functional and then we establish a key lemma on the general properties of electrostatic manifolds with boundary. In Section 3 we provide the necessary background for Theorem 1.1 and Theorem 1.2. In Section 4 we prove these uniqueness theorems. In Appendix 5 we provide some auxiliary computations needed for the purposes of the thesis.

2 Electrostatic manifolds with boundary

In this section we will provide relevant definitions and properties for electrostatic manifolds with boundary with all the necessary proofs.

2.1 Static spacetimes satisfying the source-free Einstein–Maxwell equations

We begin by showing how the source-free Einstein–Maxwell equations reduce under the static spacetime assumption and how this reduction yields the defining equations of electrostatic manifolds in general relativity. Then, we show why the obtained system of equations is equivalent to the one used in the introduction (for similar computations, see [KL18], [CLdS24] and [CB08, Section XIV.3.2]).

Definition 2.1. The *source-free Einstein–Maxwell equations* for $(n+1)$-dimensional spacetime $(\mathbb{R} \times M^n, \mathfrak{g})$ are

$$\begin{cases} \mathrm{Ric}_{\mathfrak{g}} - \frac{1}{2}\,\mathrm{R}_{\mathfrak{g}}\,\mathfrak{g} + \Lambda\mathfrak{g} = 2\left(F \circ F - \frac{1}{4}|F|^2_{\mathfrak{g}}\,\mathfrak{g}\right), & \text{(EM1)} \\ dF = 0, \quad d *_{\mathfrak{g}} F = 0, & \text{(EM2)} \end{cases}$$

where F is the *electromagnetic tensor* and $(F \circ F)_{\alpha\beta} = \mathfrak{g}^{\mu\nu} F_{\alpha\mu} F_{\beta\nu}$. The solutions of these equations are called *electro-vacuum spacetimes*.

Let us take the metric trace of (EM1):

1⌋ The left-hand side becomes

$$\begin{aligned} \mathrm{tr}_g\left(\mathrm{Ric}_{\mathfrak{g}} - \frac{1}{2}\,\mathrm{R}_{\mathfrak{g}}\,\mathfrak{g} + \Lambda\mathfrak{g}\right) &= \mathrm{R}_{\mathfrak{g}} - \frac{1}{2}\,\mathrm{R}_{\mathfrak{g}}(n+1) + \Lambda(n+1) = \\ &= -\frac{n-1}{2}\,\mathrm{R}_{\mathfrak{g}} + (n+1)\Lambda. \end{aligned}$$

2⌋ The right-hand side becomes

$$\begin{aligned} 2\left(F \circ F - \frac{1}{4}|F|^2_{\mathfrak{g}}\,\mathfrak{g}\right) &= 2\left(|F|^2_{\mathfrak{g}} - \frac{1}{4}|F|^2_{\mathfrak{g}}(n+1)\right) = \\ &= -\frac{n-3}{2}|F|^2_{\mathfrak{g}}. \end{aligned}$$

3⌋ Thus, (EM1) becomes

$$\mathrm{R}_{\mathfrak{g}} = \frac{n-3}{n-1}|F|^2_{\mathfrak{g}} + \frac{2(n+1)}{n-1}\Lambda. \tag{2.2}$$

Inserting (2.2) into (EM1) we get an alternative version of the Einstein–Maxwell equations for (n+1)-dimensional spacetime

$$\begin{cases} \mathrm{Ric}_{\mathfrak{g}} = 2\left(F \circ F - \frac{1}{2(n-1)}|F|^2_{\mathfrak{g}}\,\mathfrak{g}\right) + \frac{2}{n-1}\Lambda\mathfrak{g}, & \text{(AEM1)} \\ dF = 0, \quad d *_{\mathfrak{g}} F = 0. & \text{(AEM2)} \end{cases}$$

Definition 2.2. A *static spacetime* is a Lorentzian manifold which can be locally represented as the Lorentzian warped product of a complete Riemannian manifold (M, g) with the real line, i.e., there exists a warping function $V \in C^\infty(M^n)$ such that the Lorentzian metric takes the form $\mathfrak{g} = -V^2 dt^2 + g$ on $\mathbb{R} \times M^n$.

Remark 2.1. Thus, for a static spacetime we have the following metric decomposition:

$$\mathfrak{g}_{tt} = -V^2, \quad \mathfrak{g}_{ti} = \mathfrak{g}^{it} = 0, \quad \mathfrak{g}_{ij} = g_{ij},$$
$$\mathfrak{g}^{tt} = -\frac{1}{V^2}, \quad \mathfrak{g}^{ti} = \mathfrak{g}^{it} = 0, \quad \mathfrak{g}^{ij} = g^{ij}.$$

Measured by the static observer $\eta = V^{-1}\partial_t$ the electromagnetic tensor F can be uniquely decomposed in terms of the *electric field* E and the *magnetic field* B (see [Gou12, Section 6.4.1] for (3+1)-spacetime). We will make the typical assumption of the vanishing of the magnetic field. Then, the decomposition takes the following form:

$$\mathrm{E}^\flat = -\iota_\eta F, \quad F = \mathrm{E}^\flat \wedge \eta^\flat = V\mathrm{E}^\flat \wedge dt, \tag{2.4}$$

where ι denotes the interior multiplication on tensors, and the equations imply that $F_{ti} = -F_{it} = VE_i^\flat$ and $F_{ij} = 0$.

Let us calculate how the equations (AEM2) simplify when working in a static spacetime with electromagnetic tensor F.

1| $0 = dF = d\left(V\mathrm{E}^\flat \wedge dt\right) = d\left(V\mathrm{E}^\flat\right) \wedge dt - V\mathrm{E}^\flat \wedge ddt = d\left(V\mathrm{E}^\flat\right) \wedge dt \quad \Longrightarrow$

$$\Longrightarrow \quad d\left(V\mathrm{E}^\flat\right) = 0.$$

2| Recall that $\operatorname{div} E = *d*\mathrm{E}^\flat$. Then

2.1| Choose an orthonormal coframe $\{\varepsilon^i\}, i = \overline{1,n}$ for g and define $\varepsilon^t = Vdt$. Then, for $i, j \neq t$

$$g = \delta_{ij}\,\varepsilon^i \otimes \varepsilon^j, \quad \mathfrak{g} = -\varepsilon^t \otimes \varepsilon^t + \delta_{ij}\,\varepsilon^i \otimes \varepsilon^j.$$

2.2| Recall that $*_g(\varepsilon^{i_1} \wedge ... \wedge \varepsilon^{i_k}) = \pm\varepsilon^{j_k} \wedge ... \wedge \varepsilon^{j_{n-k}}$, where $(i_1, ..., i_k, j_1, ..., j_{n-k})$ is some permutation of $(1, 2, ..., n)$. Then

$$*_{\mathfrak{g}}(\varepsilon^{i_1} \wedge \varepsilon^t) = \pm\widehat{\varepsilon^t} \wedge \varepsilon^{j_1} \wedge ... \wedge \varepsilon^{j_{n-1}} = \pm *_g (\varepsilon^{i_1}),$$

where $\widehat{...}$ denotes the omitted term.

2.3| Thus,

$$0 = d *_{\mathfrak{g}} F = d *_{\mathfrak{g}} \left(V\mathrm{E}^\flat \wedge dt\right) = d *_{\mathfrak{g}} \left(V\mathrm{E}^\flat \wedge \frac{\varepsilon^t}{V}\right) = d *_{\mathfrak{g}} \left(\mathrm{E}^\flat \wedge \varepsilon^t\right) =$$
$$= \pm d *_g \mathrm{E}^\flat \quad \Longrightarrow \quad \operatorname{div}_g E = 0.$$

Therefore, the equations EM2 are reduced to

$$\operatorname{div}_g E = 0, \quad d\left(V\mathrm{E}^\flat\right) = 0, \tag{2.5}$$

where for $n = 3$ case we have $d\left(V\mathrm{E}^\flat\right) = \operatorname{curl}(VE) = 0$.

Next, we decompose (EM1) in the same setting. We start by writing the standard formulae from Riemannian geometry which we will use for calculations (see for example [Lee18]).

$$\Gamma^k_{ij} = \frac{1}{2} g^{kl} \left(\partial_i g_{jl} - \partial_l g_{ij} + \partial_j g_{il} \right), \tag{2.6}$$

$$\Gamma^k_{ik} = \frac{1}{2} g^{kl} \partial_i g_{kl}, \tag{2.7}$$

$$R_{ki,j}{}^{l} = \partial_k \Gamma^l_{ij} - \partial_i \Gamma^l_{kj} + \Gamma^m_{ij} \Gamma^l_{km} - \Gamma^m_{kj} \Gamma^l_{im}, \tag{2.8}$$

$$R_{ij} = R_{ki,j}{}^{k} = \partial_k \Gamma^k_{ij} - \partial_i \Gamma^k_{kj} + \Gamma^m_{ij} \Gamma^k_{km} - \Gamma^m_{kj} \Gamma^k_{im}, \tag{2.9}$$

$$\left(\mathrm{Hess}_g V \right)_{ij} = \partial_i \partial_j V - \Gamma^k_{ij} \partial_k V, \tag{2.10}$$

$$\Delta_g V = g^{ij} \left(\partial_i \partial_j V - \Gamma^k_{ij} \partial_k V \right) = \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j V \right), \tag{2.11}$$

where $|g| \coloneqq |\det (g_{ij})|$.

Remark 2.2. For the remainder of this section, we use t to denote temporal components, other English letters to denote spatial components, and Greek letters when the nature of the component is not specified. Also, $\widetilde{\Gamma}_{\mu\nu}$ (resp. Γ_{ij}) are Christoffel symbols of $\mathfrak{g}$ (resp. g) and $\widetilde{R}_{\mu\nu}$ (resp. R_{ij}) are components of the Ricci tensor of $\mathfrak{g}$ (resp. g).

Using the aforementioned formulae, we calculate the Christoffel symbols of $\mathfrak{g}$.

- $\widetilde{\Gamma}^t_{ti} = \frac{1}{2} \mathfrak{g}^{t\mu} \left(\partial_t \mathfrak{g}_{it} - \partial_\mu \mathfrak{g}_{ti} + \partial_i \mathfrak{g}_{t\mu} \right) = \frac{1}{2} \mathfrak{g}^{tt} \left(\partial_t \mathfrak{g}_{it} - \partial_t \mathfrak{g}_{ti} + \partial_i \mathfrak{g}_{tt} \right) =$

  $= \frac{1}{2} \mathfrak{g}^{tt} \partial_i \mathfrak{g}_{tt} = \frac{1}{2} \frac{1}{V^2} 2V \partial_i V = \frac{\partial_i V}{V},$

- $\widetilde{\Gamma}^i_{tt} = \frac{1}{2} \mathfrak{g}^{i\mu} \left(\partial_t \mathfrak{g}_{t\mu} - \partial_\mu \mathfrak{g}_{tt} + \partial_t \mathfrak{g}_{t\mu} \right) = \frac{1}{2} \mathfrak{g}^{ij} \left(\partial_t \mathfrak{g}_{tj} - \partial_j \mathfrak{g}_{tt} + \partial_t \mathfrak{g}_{tj} \right) =$

  $= \frac{1}{2} \mathfrak{g}^{ij} \left(-\partial_j \mathfrak{g}_{tt} \right) = \frac{1}{2} g^{ij} 2V \partial_j V = V g^{ij} \partial_j V,$

- $\widetilde{\Gamma}^t_{ij} = \frac{1}{2} \mathfrak{g}^{t\mu} \left(\partial_i \mathfrak{g}_{j\mu} - \partial_\mu \mathfrak{g}_{ij} + \partial_j \mathfrak{g}_{i\mu} \right) = \frac{1}{2} \mathfrak{g}^{tt} \left(\partial_i \mathfrak{g}_{jt} - \partial_t \mathfrak{g}_{ij} + \partial_j \mathfrak{g}_{it} \right) = 0,$

- $\widetilde{\Gamma}^i_{tj} = \frac{1}{2} \mathfrak{g}^{i\mu} \left(\partial_t \mathfrak{g}_{j\mu} - \partial_\mu \mathfrak{g}_{tj} + \partial_j \mathfrak{g}_{t\mu} \right) = \frac{1}{2} \mathfrak{g}^{ij} \left(\partial_t \mathfrak{g}_{jj} - \partial_j \mathfrak{g}_{tj} + \partial_j \mathfrak{g}_{tj} \right) = 0,$

- $\widetilde{\Gamma}^t_{tt} = \frac{1}{2} \mathfrak{g}^{t\mu} \left(\partial_t \mathfrak{g}_{t\mu} - \partial_\mu \mathfrak{g}_{tt} + \partial_t \mathfrak{g}_{t\mu} \right) = \frac{1}{2} \mathfrak{g}^{tt} \left(\partial_t \mathfrak{g}_{tt} - \partial_t \mathfrak{g}_{tt} + \partial_t \mathfrak{g}_{tt} \right) = \frac{1}{2} \mathfrak{g}^{tt} \partial_t \mathfrak{g}_{tt} = 0.$

Thus, we have

$$\widetilde{\Gamma}^t_{ti} = \frac{\partial_i V}{V}, \quad \widetilde{\Gamma}^i_{tt} = V g^{ij} \partial_j V, \quad \widetilde{\Gamma}^t_{ij} = \widetilde{\Gamma}^i_{tj} = \widetilde{\Gamma}^t_{tt} = 0, \quad \widetilde{\Gamma}^k_{ij} = \Gamma^k_{ij}, \tag{2.12}$$

where $\widetilde{\Gamma}$ and Γ denote Christoffel symbols of $\mathfrak{g}$ and g respectively.

Then, let us notice that

$$\Delta_g V = \frac{1}{\sqrt{|g|}} \partial_i \left(\sqrt{|g|} g^{ij} \partial_j V \right) = \partial_i \left(g^{ij} \partial_j V \right) + \frac{\partial_i \sqrt{|g|}}{\sqrt{|g|}} \left(g^{ij} \partial_j V \right) =$$

$$= \partial_i \left(g^{ij} \partial_j V \right) + \partial_i \log \sqrt{|g|} \left(g^{ij} \partial_j V \right) = \partial_i \left(g^{ij} \partial_j V \right) + \Gamma^k_{ik} \left(g^{ij} \partial_j V \right), \tag{2.13}$$

where we used (2.11), (2.7) and Jacobi's formula $\frac{d}{dx}\det A = \det A \operatorname{tr}\left(A^{-1}\frac{dA}{dx}\right)$ for $A = (g_{ij})$.

Now we are ready to calculate the components of the Ricci tensor $\mathrm{Ric}_{\mathfrak{g}}$.

- Ricci tensor temporal component:

$$\begin{aligned}
\widetilde{R}_{tt} &= \partial_\alpha \widetilde{\Gamma}^\alpha_{tt} - \partial_t \widetilde{\Gamma}^\alpha_{\alpha t} + \widetilde{\Gamma}^\beta_{tt}\widetilde{\Gamma}^\alpha_{\alpha\beta} - \widetilde{\Gamma}^\beta_{\alpha t}\widetilde{\Gamma}^\alpha_{t\beta} = \\
&= \partial_k \widetilde{\Gamma}^k_{tt} - 0 + \left(\widetilde{\Gamma}^m_{tt}\Gamma^k_{km} + \widetilde{\Gamma}^m_{tt}\Gamma^t_{tm}\right) - \left(\widetilde{\Gamma}^t_{kt}\widetilde{\Gamma}^k_{tt} + \widetilde{\Gamma}^m_{tt}\widetilde{\Gamma}^t_{tm}\right) = \\
&= \partial_k\left(V g^{kj}\partial_j V\right) + V\Gamma^k_{km}(g^{mj}\partial_j V) - \widetilde{\Gamma}^t_{kt}\widetilde{\Gamma}^k_{tt} = \\
&= (\partial_k V)(g^{kj}\partial_j V) + V\partial_k\left(g^{kj}\partial_j V\right) + V\Gamma^k_{km}(g^{mj}\partial_j V) - (\partial_k V)(g^{kj}\partial_j V) = \\
&= V\partial_k\left(g^{kj}\partial_j V\right) + V\Gamma^k_{km}(g^{mj}\partial_j V) = \\
&= V\Delta_g V.
\end{aligned}$$

- Ricci tensor spatial components $\widetilde{R}_{ij}$:

$$\begin{aligned}
\widetilde{R}_{ij} &= \partial_\alpha \widetilde{\Gamma}^\alpha_{ij} - \partial_i \widetilde{\Gamma}^\alpha_{\alpha j} + \widetilde{\Gamma}^\beta_{ij}\widetilde{\Gamma}^\alpha_{\alpha\beta} - \widetilde{\Gamma}^\beta_{\alpha j}\widetilde{\Gamma}^\alpha_{i\beta} = \\
&= \partial_k \Gamma^k_{ij} - \partial_i\left(\widetilde{\Gamma}^t_{tj} + \Gamma^k_{kj}\right) + \left(\Gamma^m_{ij}\widetilde{\Gamma}^t_{tm} + \Gamma^m_{ij}\Gamma^k_{km}\right) - \left(\widetilde{\Gamma}^t_{tj}\widetilde{\Gamma}^t_{it} + \Gamma^m_{kj}\Gamma^k_{im}\right) = \\
&= \partial_k \Gamma^k_{ij} - \left(\frac{V\partial_i\partial_j V - (\partial_i V)(\partial_j V)}{V^2} + \partial_i \Gamma^k_{kj}\right) + \\
&+ \left(\Gamma^m_{ij}\frac{\partial_m V}{V} + \Gamma^m_{ij}\Gamma^k_{km}\right) - \left(\widetilde{\Gamma}^t_{tj}\widetilde{\Gamma}^t_{it} + \Gamma^m_{kj}\Gamma^k_{im}\right) = \\
&= R_{ij} - \frac{\partial_i\partial_j V}{V} + \frac{(\partial_i V)(\partial_j V)}{V^2} + \Gamma^m_{ij}\frac{\partial_m V}{V} - \frac{(\partial_i V)(\partial_j V)}{V^2} = \\
&= R_{ij} - \frac{1}{V}\left(\mathrm{Hess}_g V\right)_{ij}.
\end{aligned}$$

- Ricci tensor mixed components $\widetilde{R}_{tj} = \widetilde{R}_{jt}$:

$$\begin{aligned}
\widetilde{R}_{tj} &= \partial_\alpha \widetilde{\Gamma}^\alpha_{tj} - \partial_t \widetilde{\Gamma}^\alpha_{\alpha j} + \widetilde{\Gamma}^\beta_{tj}\widetilde{\Gamma}^\alpha_{\alpha\beta} - \widetilde{\Gamma}^\beta_{\alpha j}\widetilde{\Gamma}^\alpha_{t\beta} = \\
&= \partial_t \widetilde{\Gamma}^t_{tj} - \partial_t \widetilde{\Gamma}^t_{tj} = \\
&= 0.
\end{aligned}$$

Therefore, $\mathrm{Ric}_{\mathfrak{g}}$ can be decomposed into

$$\widetilde{R}_{tt} = V\Delta_g V, \quad \widetilde{R}_{tj} = \widetilde{R}_{jt} = 0, \quad \widetilde{R}_{ij} = R_{ij} - \frac{1}{V}\left(\mathrm{Hess}_g V\right)_{ij}. \tag{2.14}$$

Using (2.4) and 2.1, the right-hand side of (EM1) with respect to the static spacetime metric can be simplified in the following way:

$$(F\circ F)_{\alpha\beta}=\mathfrak{g}^{\mu\nu}F_{\mu\alpha}F_{\nu\beta}=\mathfrak{g}^{tt}F_{t\alpha}F_{t\beta}+\mathfrak{g}^{ij}F_{i\alpha}F_{j\beta}=-\frac{1}{V^2}F_{t\alpha}F_{t\beta}+g^{ij}F_{i\alpha}F_{j\beta}, \tag{2.15}$$

$$\begin{aligned}F_{\mu\nu}F^{\mu\nu}&=F_{ti}F^{ti}+F_{it}F^{it}=2F_{ti}F^{ti}=2\left(V\mathrm{E}^\flat{}_i\right)\left(\mathfrak{g}^{tt}\mathfrak{g}^{ij}F_{tj}\right)=\\&=2\left(V\mathrm{E}^\flat{}_i\right)\left(-\frac{1}{V}g^{ij}\mathrm{E}^\flat{}_j\right)=-2g^{ij}\mathrm{E}^\flat{}_i\mathrm{E}^\flat{}_j.\end{aligned} \tag{2.16}$$

Thus, with (2.16) and (2.15)

$$(F\circ F)_{\alpha\beta}-\frac{1}{2(n-1)}F_{\mu\nu}F^{\mu\nu}\,\mathfrak{g}_{\alpha\beta}=-\frac{1}{V^2}F_{t\alpha}F_{t\beta}+g^{ij}F_{i\alpha}F_{j\beta}+\frac{1}{n-1}\frac{1}{V^2}g^{ij}\mathrm{E}^\flat{}_i\mathrm{E}^\flat{}_j\,\mathfrak{g}_{\alpha\beta}. \tag{2.17}$$

Finally, the combination of (2.14), (2.17) and (2.5) implies that in a static spacetime with aforementioned electromagnetic tensor $F=V\mathrm{E}^\flat\wedge dt$, the Einstein–Maxwell equations (AEM1), (AEM2) reduce to

$$R_{ij}-\frac{1}{V}\left(\mathrm{Hess}_g V\right)_{ij}=2\left(-\mathrm{E}^\flat{}_i\mathrm{E}^\flat{}_j+\frac{1}{n-1}g^{ij}\mathrm{E}^\flat{}_i\mathrm{E}^\flat{}_j\,g_{ij}\right)+\frac{2}{n-1}\Lambda g_{ij}, \tag{2.18}$$

$$\begin{aligned}V\Delta_g V&=2\left(V^2|\mathrm{E}|_g^2-\frac{1}{n-1}V^2|\mathrm{E}|_g^2\right)-V^2\frac{2}{n-1}\Lambda=\\&=V^2\left(\frac{2(n-2)}{n-1}|\mathrm{E}|_g^2-\frac{2}{n-1}\Lambda\right),\end{aligned} \tag{2.19}$$

$$\mathrm{div}_g E=0,\quad d\left(V\mathrm{E}^\flat\right)=0. \tag{2.20}$$

Putting (2.18), (2.19), (2.20) together and imposing natural Robin boundary condition, we obtain the following class of manifolds:

Definition 2.3. *Electrostatic manifold with boundary* is a quadruple (M^n,g,V,E), where (M^n,g) is a Riemannian manifold with boundary, which admits a nontrivial smooth solution $V:M\to\mathbb{R}$, called the *static potential*, to the following system

$$\begin{cases}\mathrm{Hess}_g V=V\left(\mathrm{Ric}_g-\frac{2}{n-1}\Lambda g+2\mathrm{E}^\flat\otimes\mathrm{E}^\flat-\frac{2}{n-1}|\mathrm{E}|_g^2\,g\right) & \text{in } M, \quad \text{(E1)}\\ \Delta_g V=V\left(\mathfrak{C}_n^2|\mathrm{E}|_g^2-\frac{2}{n-1}\Lambda\right) & \text{in } M, \quad \text{(E2)}\\ \mathrm{div}_g E=0,\quad d\left(V\mathrm{E}^\flat\right)=0 & \text{in } M, \quad \text{(E3)}\\ \frac{\partial V}{\partial\nu}g-VB_g=0 & \text{on } \partial M, \quad \text{(E4)}\end{cases}$$

where $E\in\Gamma(\mathrm{T}M)$ is a tangent vector field called *electric field*, $\mathrm{E}^\flat=g(E,\cdot)\in\Gamma(\mathrm{T}^*M)$ is the one-form, metrically dual to E, B_g is the second fundamental form of ∂M with respect to the outward unit normal vector field ν, and $\mathfrak{C}_n:=\sqrt{2(n-2)/(n-1)}$.

By taking the metric trace of (E1) and (E4), we additionally get

$$\Delta_g V = V\left(\mathrm{R}_g - \frac{2n}{n-1}\Lambda - \frac{2}{n-1}|\mathrm{E}|_g^2\right) \qquad \text{in } M, \tag{TE1}$$

$$\frac{\partial V}{\partial \nu} = \frac{H_g}{n-1}V \qquad \text{on } \partial M. \tag{TE2}$$

Thus, (TE1),(TE2), (E2) and (E4) imply that if $V \not\equiv 0$ on an electrostatic manifold (with boundary) (M, g, V, E), then the following equations hold:

$$\mathrm{R}_g = 2|\mathrm{E}|_g^2 + 2\Lambda \qquad \text{in } M, \tag{NE1}$$

$$B_g = \frac{H_g}{n-1}g \qquad \text{on } \partial M. \tag{NE2}$$

Remark 2.3. If $V \not\equiv 0$, then the equation (NE1) is indeed true not only in $M \setminus V^{-1}(0)$ but in all M. We know that $V^{-1}(0)$ is an embedded hypersurface (see (a) in Lemma 2.8 below) and so it is a regular level set of a smooth function V with 0 having measure zero in $\mathbb{R}$. Then, $V^{-1}(0)$ has measure zero in M and, hence, $M \setminus V^{-1}(0)$ is dense in M (see, for example, [Lee12, Proposition 6.8]) and therefore the smooth function $F := \mathrm{R}_g - 2|\mathrm{E}|_g^2 - 2\Lambda$ must be equal to zero in the closure of $M \setminus V^{-1}(0)$ due to continuity. In particular, if $F(x) = c \neq 0$ for some $x \in V^{-1}(0)$, then, due to the fact that manifolds are Hausdorff, in $\mathbb{R}$ there exist open neighborhoods $U_0 \ni 0$ and $U \ni c$ such that $U \cap U_0 = \varnothing$ and, hence, $\left(F^{-1}(U_0) = M \setminus V^{-1}(0)\right) \cap F^{-1}(U) = \varnothing$. However, since $M \setminus V^{-1}(0)$ is dense in M, then $F^{-1}(U)$ must intersect $M \setminus V^{-1}(0)$ which contradicts $F^{-1}(U_0) \cap F^{-1}(U) = \varnothing$.

Equation (NE2) is also true in all M as $V^{-1}(0)$ is totally geodesic and therefore both B_g and H_g are zero in $V^{-1}(0) \cap \partial M$ if there is such intersection.

2.2 General properties of electrostatic manifolds with boundary

Now we take a closer look at the general properties of electrostatic manifolds with boundary, showing how Definition 2.3 is equivalent to the one we used in the Introduction along the way. We start by rewriting [CLdS24, Proposition 5] and showing that it is applicable to n-dimensional electrostatic manifolds (with boundary).

Lemma 2.4. Let (M, g, V, E) be electrostatic manifold (with boundary), then the equations (E1), (E2) and (E3) are equivalent to the following second order overdetermined elliptic equation:

$$\mathrm{Hess}_g V - (\Delta_g V)\, g - V\,\mathrm{Ric}_g = 2V\left(\mathrm{E}^\flat \otimes \mathrm{E}^\flat - |\mathrm{E}|_g^2\, g\right). \tag{2.22}$$

In particular, for the formal L^2-adjoint operator to the linearized scalar curvature operator $D\,\mathrm{R}^*\big|_g(V) = \mathrm{Hess}_g V - (\Delta_g V)\, g - V\,\mathrm{Ric}_g$, we see that $V \in \mathrm{Ker}\left(D\,\mathrm{R}^*\big|_g\right)$ if, and only if, $E \equiv 0$.

Proof. The equations (E1) and (E2) imply (2.22):

$$\begin{aligned}\operatorname{Hess}_g V &= V \operatorname{Ric}_g + 2V \mathrm{E}^\flat \otimes \mathrm{E}^\flat - V \frac{2}{n-1} |\mathrm{E}|_g^2 g + \left(\Delta_g V - V \mathfrak{C}_n^2 |\mathrm{E}|_g^2\right) g = \\ &= V \operatorname{Ric}_g + 2V \mathrm{E}^\flat \otimes \mathrm{E}^\flat + (\Delta_g V)\, g - 2V |\mathrm{E}|_g^2 \left(\frac{n-2}{n-1} + \frac{1}{n-1}\right) g = \\ &= V \operatorname{Ric}_g + (\Delta_g V)\, g + 2V \left(\mathrm{E}^\flat \otimes \mathrm{E}^\flat - |\mathrm{E}|_g^2\, g\right).\end{aligned}$$

For the proof of the converse implication, see the proof of [CLdS24, Proposition 5] where the condition $d\left(V \mathrm{E}^\flat\right) = 0$ from (E3) is used from the beginning.

□

With Lemma 2.4 we obtain the definition of electrostatic manifolds with boundary which we used in the introduction.

Corollary 2.5. *Electrostatic manifold with boundary* is a quadruple (M^n, g, V, E), where (M^n, g) is a Riemannian manifold with boundary, which admits a nontrivial smooth solution $V : M \to \mathbb{R}$, called the *static potential*, to the following system

$$\begin{cases} \operatorname{Hess}_g V - (\Delta_g V)\, g - V \operatorname{Ric}_g = 2V \left(\mathrm{E}^\flat \otimes \mathrm{E}^\flat - |\mathrm{E}|_g^2\, g\right) & \text{in } M, \qquad \text{(AE1)} \\ \dfrac{\partial V}{\partial \nu} g - V B_g = 0 & \text{on } \partial M. \qquad \text{(AE2)} \end{cases}$$

By taking the metric trace of the equation (AE1) we additionally get

$$\Delta_g V = V \left(2|\mathrm{E}|_g^2 - \frac{1}{n-1} \mathrm{R}_g\right).$$

Now we take a look at electrostatic manifolds with boundary from the perspective characteristic to the prescribed scalar curvature problems.

Theorem 2.6. The quadruple (M^n, g_o, V, E) is an electrostatic manifold with boundary if, and only if, $g_o \in \mathcal{R}(M)$ is critical for the functional

$$\mathcal{F}[g] = \int_M V \left(\mathrm{R}_g - 6|\mathrm{E}|_g^2\right) dv_{g_o} + 4 \int_M V |\mathrm{E}|_g^2\, dv_g + 2 \int_{\partial M} V H_g\, ds_{g_o}, \tag{2.24}$$

where $\mathcal{R}(M)$ is the set of Riemannian metrics on M.

Proof.

1] Let δ_g denote the variation with respect to the metric g and $h = \left.\dfrac{d}{dt}\right|_{t=0} g(t)$, $g(0) = g$. Then

$$\begin{aligned}\delta_g \left(|\mathrm{E}|_g^2\right)(h) = \delta_g \left(g_{ij} E^i E^j\right) &= \left.\frac{d}{dt}\right|_{t=0} \left(g(t)_{ij} E^i E^j\right) = \\ &= h_{ij} E^i E^j = g^{ik} g^{jl} \mathrm{E}^\flat{}_k \mathrm{E}^\flat{}_l h_{ij} = \\ &= \langle \mathrm{E}^\flat \otimes \mathrm{E}^\flat, h \rangle_g. \end{aligned} \tag{2.25}$$

2| Using the Leibniz rule we obtain

$$\delta_g\left(|\mathrm{E}|_g^2\, dv_g\right)(h) = \delta_g\left(|\mathrm{E}|_g^2\right)(h)\, dv_g + |\mathrm{E}|_g^2\, \delta_g\left(\, dv_g\right)(h) =$$

$$= \langle \mathrm{E}^\flat \otimes \mathrm{E}^\flat + \frac{1}{2}|\mathrm{E}|_g^2\, g, h\rangle_g\, dv_g, \tag{2.26}$$

where we used $\delta_g(\, dv_g)(h) = \frac{1}{2}\langle g, h\rangle_g\, dv_g$.

3| Also notice that using (2.25) and (2.26) we get

$$\left\langle -2V\mathrm{E}^\flat \otimes \mathrm{E}^\flat + 2V|\mathrm{E}|_g^2\, g, h\right\rangle_g\, dv_g = V\left(\delta_g\left(-6|\mathrm{E}|_g^2\right)(h)\, dv_g + 4\delta_g\left(|\mathrm{E}|_g^2\, dv_g\right)\right). \tag{2.27}$$

4| Let $\gamma := g_{\partial M}, \gamma_o := (g_o)_{\partial M}$ and recall that $g_o \in \mathcal{R}(M)$ is critical for the functional $\mathcal{F}[g]$ if, and only if, $\delta_g\mathcal{F}(g_o)(h) = 0$. Let us calculate $\delta_g\mathcal{F}(g_o)(h)$.

5| Variation of the first term in (2.24):

$$\delta_g\left(\int_M V\left(\mathrm{R}_g - 6|\mathrm{E}|_g^2\right)\, dv_{g_o}\right)(h) = \int_M V\delta_g(\mathrm{R}_g\, dv_g)(h) - 6\int_M V\delta_g(|\mathrm{E}|_g^2\, dv_{g_o})(h) =$$

$$= \int_M V\left(\delta_g\, \mathrm{R}_g\, h\right)\, dv_{g_o} - 6\int_M V\langle \mathrm{E}^\flat \otimes \mathrm{E}^\flat, h\rangle_g\, dv_{g_o}.$$

6| Variation of the second term in (2.24):

$$\delta_g\left(4\int_M V|\mathrm{E}|_g^2\, dv_g\right)(h) = 4\int_M V\langle \mathrm{E}^\flat \otimes \mathrm{E}^\flat + \frac{1}{2}|\mathrm{E}|_g^2\, g, h\rangle_g\, dv_g.$$

7| Variation of the third term in (2.24):

$$\delta_g\left(2\int_{\partial M} V H_\gamma\, ds_{\gamma_o}\right)(h) = 2\int_{\partial M} V\left(\delta_\gamma H_\gamma h\right) ds_{\gamma_o}.$$

8| Summing these terms, using the calculations after Lemma 2.1 in [CV19] and equations

(2.25), (2.26), we obtain

$$\delta_g\mathcal{F}(g)(h) = \int_M V\left(\delta_g \mathrm{R}_g\, h\right)\, dv_{g_o} - 6\int_M V\langle \mathrm{E}^\flat\otimes\mathrm{E}^\flat, h\rangle_g\, dv_{g_o} +$$

$$+\,4\int_M V\langle \mathrm{E}^\flat\otimes\mathrm{E}^\flat + \frac{1}{2}|\mathrm{E}|_g^2\, g, h\rangle_g\, dv_g + 2\int_{\partial M} V\left(\delta_g H_g h\right) ds_{\gamma_o} =$$

$$= \int_M V\delta_g \mathrm{R}_g\, h\, dv_{g_o} + 2\int_{\partial M} V\delta_\gamma H_\gamma h\, ds_{\gamma_o} - \int_M \langle 2V\mathrm{E}^\flat\otimes\mathrm{E}^\flat + 2V|\mathrm{E}|_g^2\, g, h\rangle_g\, dv_{g_o} =$$

$$= \int_M \left(-\Delta_g V g(\mathrm{tr}_g\, h) + \langle \mathrm{Hess}_g\, V, h\rangle_g - V\,\langle h, \mathrm{Ric}_g\rangle\right)\, dv_{g_o} +$$

$$+\int_{\partial M}\left(\mathrm{tr}_\gamma\, h\frac{\partial V}{\partial\nu} - V\,\langle B_\gamma, h\rangle_\gamma\right)\, ds_{\gamma_o} - \int_M \langle 2V\mathrm{E}^\flat\otimes\mathrm{E}^\flat + 2V|\mathrm{E}|_g^2\, g, h\rangle_g\, dv_{g_o} =$$

$$= \int_M \left\langle (-\Delta_g V)g + \mathrm{Hess}_g\, V - V\,\mathrm{Ric}_g - 2V\mathrm{E}^\flat\otimes\mathrm{E}^\flat + 2V|\mathrm{E}|_g^2\, g, h\right\rangle_g\, dv_{g_o} +$$

$$+\int_{\partial M}\left\langle \frac{\partial V}{\partial\nu} g - VB_\gamma, h\right\rangle_\gamma\, ds_{\gamma_o},$$

where we used that $\mathrm{tr}_g\, h = \langle g, h\rangle_g$ and without loss of generality we added the assumption of working with g_o for the volume form in the last integral in the third line.

9| Since $\delta_g\mathcal{F}(g_o)(h)$ must vanish for any arbitrary variation h, g_o is a critical point of $\mathcal{F}[g]$ if, and only if,

$$\begin{cases} \mathrm{Hess}_g\, V - (\Delta_g V)\, g - V\,\mathrm{Ric}_g = 2V\left(\mathrm{E}^\flat\otimes\mathrm{E}^\flat - |\mathrm{E}|_g^2\, g\right) & \text{in } M, \\ \dfrac{\partial V}{\partial\nu} g - VB_g = 0 & \text{on } \partial M, \end{cases}$$

which implies that (M, g_o, V, E) is an electrostatic manifold with boundary by Corollary 2.5.

□

Proposition 2.7. A nontrivial $V\in C^\infty(M)$ is an *electrostatic potential* if, and only if, it satisfies

$$\left(D\left(\mathrm{R}_g - 2|\mathrm{E}|_g^2, 2H_g\right)\right)^* = \left(-2V|\mathrm{E}|_g^2\, g, 0\right). \tag{2.29}$$

where the left-hand side of (2.29) is the formal L^2-adjoint linearization of $\left(\mathrm{R}_g - 2|\mathrm{E}|_g^2, 2H_g\right)$. In particular, $V\in\mathrm{Ker}\left(D\left(\mathrm{R}_g, 2H_g\right)\right)^* \iff E\equiv 0$.

Proof. Consider the following functional on $\mathcal{R}(M)$: $g\mapsto\left(\mathrm{R}_g - 2|\mathrm{E}|_g^2, 2H_g\right)$. From (2.25) and [CV19] follows that the L^2-adjoint for the linearization of this functional is

$$\left(D\left(\mathrm{R}_g - 2|\mathrm{E}|_g^2, 2H_g\right)\right)^* = \left(\mathrm{Hess}_g\, V - (\Delta_g V)\, g - V\,\mathrm{Ric}_g - 2V\mathrm{E}^\flat\otimes\mathrm{E}^\flat, \frac{\partial V}{\partial\nu} g - VB_g\right).$$

Thus, the statement of the Proposition 2.7 follows.

□

The next lemma on the general properties of electrostatic manifolds with boundary follows from [CN23, Proposition 1], [CLdS24, Lemma 4] and the equations for electrostatic manifolds with boundary derived earlier in the paper.

Lemma 2.8. Let (M^n, g, V, E) be an electrostatic manifold with boundary and $\Sigma = V^{-1}(0)$ nonempty. Then:

(a) Σ is an embedded totally geodesic hypersurface in M. More precisely:

(a.1) If $\Sigma \cap \partial M = \emptyset$, then Σ is a totally geodesic hypersurface contained in the interior of M, $\operatorname{Int} M$;

(a.2) If $\Sigma \cap \partial M \neq \emptyset$, then each connected component Σ_0 of Σ such that $\Sigma_0 \cap \partial M \neq \emptyset$ is a free boundary totally geodesic hypersurface of M. In particular, $\partial\Sigma_0$ is a totally geodesic hypersurface of ∂M.

(b) The surface gravity $\kappa := |\nabla^g V|$ is a positive constant on each connected component of Σ;

(c) The electric field E and $\nabla^g V$ are linearly dependent along Σ;

(d) The scalar curvature $\mathrm{R}_g = 2|\mathrm{E}|_g^2 + 2\Lambda$;

(e) On ∂M, we have

$$dH_g = 2E^\flat(\nu)E^\flat, \tag{2.30}$$

where ν is the unit normal vector field to ∂M. In particular, for the case when $E \in \Gamma(T\partial M)$ or $E \in \Gamma(N\partial M)$ (from which the latter case, for example, naturally occurs for spherically symmetric spaces) we have

$$dH_g = 0 \text{ and } \operatorname{Ric}_g(\nu, X) = 0;$$

(f) The boundary ∂M is totally umbilical. In particular, for the case where $E \in \Gamma(T\partial M)$ or $E \in \Gamma(N\partial M)$ the mean curvature of the boundary H_g is constant on each connected component of ∂M and, additionally, from item (a.2) it follows that if Σ_0 is a connected component of Σ such that $\Sigma_0 \cap \partial M \neq \emptyset$, the mean curvature of $\partial\Sigma_0$ in Σ_0 is locally constant.

Proof.

1. Items (a), (b), and (c) follow from [CN23, Proposition 1], [CLdS24, Lemma 4] and [Aro57].

2. Items (d) and (f) follow from (NE1) and (NE2) respectively (also see Remark 2.3).

3. As in the proof of [CN23, Proposition 1], we consider $p \in \partial M, X \in \mathrm{T}_p\partial M, \nu \in \mathrm{N}_p\partial M$. Then, recalling that $B_g(X, Y) = g(\nabla_X \nu, Y) \quad \forall X, Y \in \mathrm{T}_p\partial M$, at $p \in \partial M$ we have

$$\begin{aligned}
\operatorname{Hess}_g V(X, \nu) &= \nabla_X \nabla_\nu V - \nabla_{\nabla_X \nu} V = X(\nu(V)) - (\nabla_X \nu)(V) = \\
&= X\left(\frac{\partial V}{\partial \nu}\right) - g\big(\nabla^M V, \nabla_X \nu\big) = X\left(\frac{\partial V}{\partial \nu}\right) - B_g\big(\nabla^{\partial M} V, X\big) = \\
&= X\left(\frac{H_g}{n-1} V\right) - \frac{H_g}{n-1} X(V) = \\
&= \frac{X(H_g)}{n-1} V,
\end{aligned} \tag{2.31}$$

where we used (TE2) and (NE2) for the penultimate equality, and the Leibniz rule for the ultimate equality.

Recall the contracted Codazzi equation:

$$\operatorname{div}_{\partial M} B_g - dH_g = \operatorname{Ric}_g(\nu, \cdot). \tag{2.32}$$

Using the formula for the divergence of a $(0,2)$-tensor with respect to an orthonormal frame $\{e_i\}$ on $\mathrm{T}_p\partial M$ we get

$$\begin{aligned}
(\operatorname{div}_{\partial M} B_g)(X) &= \sum_{i=1}^{n-1} \big(\nabla^{\partial M}_{e_i} B_g\big)(e_i, X) = \\
&= \sum_{i=1}^{n-1} \big(e_i(B_g(e_i, X)) - B_g\big(\nabla^{\partial M}_{e_i} e_i, X\big) - B_g\big(e_i, \nabla^{\partial M}_{e_i} X\big)\big) = \\
&= \sum_{i=1}^{n-1} \big(e_i(B_g(e_i, X)) - B_g\big(e_i, \nabla^{\partial M}_{e_i} X\big)\big) = \\
&= \sum_{i=1}^{n-1} \left(e_i\left(\frac{H_g}{n-1} g(e_i, X)\right) - \frac{H_g}{n-1} g\big(e_i, \nabla^{\partial M}_{e_i} X\big)\right) = \\
&= \sum_{i=1}^{n-1} \left(\frac{e_i(H_g)}{n-1} g(e_i, X) + \frac{H_g}{n-1}\Big(e_i(g(e_i, X)) - g\big(e_i, \nabla^{\partial M}_{e_i} X\big)\Big)\right) = \\
&= \sum_{i=1}^{n-1} \left(\frac{e_i(H_g)}{n-1} g(e_i, X)\right) = \\
&= \frac{X(H_g)}{n-1},
\end{aligned} \tag{2.33}$$

where we used the fact that $\nabla^{\partial M}_{e_i} e_i = 0$ and $\nabla g = 0$ for the Levi–Civita connection.

Then, the equations (2.32), (2.31) and (2.33) imply

$$\begin{aligned}
\operatorname{Ric}_g(\nu, X) &= (\operatorname{div}_{\partial M} B_g)(X) - dH_g(X) = \\
&= \frac{X(H_g)}{n-1} - X(H_g) = \\
&= -\frac{n-2}{n-1} X(H_g).
\end{aligned} \tag{2.34}$$

Finally, substituting (2.31) and (2.34) into (E1) we obtain

$$X(H_g) = 2\mathrm{E}^\flat(\nu)\mathrm{E}^\flat(X), \tag{2.35}$$

which is exactly the equation (2.30).

□

3 Preliminaries for the uniqueness theorems

In this section, we provide definitions and obtain results that are instrumental in both the statement and the proofs of Theorem 1.1 and Theorem 1.2.

When working with asymptotically flat spacetime, as in [KL18] it is also natural to assume that the domain of outer communication for the static spacetime defined in Section 2.1 is globally hyperbolic. Then, as the authors of the aforementioned article note, topological censorship (see [FSW93, FSW95, CW94, Gal95]) implies that M is simply connected and, due to $d(VE^\flat) = 0$ (that is, $VE^\flat$ being closed) from (E3), we deduce that $VE^\flat$ must also be exact, and so there exists a globally defined smooth function $\Psi : M \to \mathbb{R}$ called *electric potential* such that $VE = -\nabla^g \Psi$.

However, the only assumption we require is the exactness of $VE^\flat$ and, consequently, the existence of the aforementioned electric potential Ψ. We make no other a priori assumptions: the domain of outer communication need not be globally hyperbolic, nor must an asymptotically flat electrostatic manifold with boundary be simply connected.

3.1 Reissner–Nordström manifold

Now we introduce Reissner-Norström manifolds. Define the *static potential* $V_{m,q}$ and the *electric field* $E_{m,q}$ in the following way:

$$V_{m,q}(r) = \sqrt{1 - \frac{2m}{r^{n-2}} + \frac{q^2}{r^{2(n-2)}}}, \quad E_{m,q}(r) = \frac{n-2}{\mathfrak{C}_n} \frac{q}{r^{n-1}} V_{m,q}(r) \partial_r.$$

Then, the $(n+1)$-dimensional Reissner–Nordström spacetime of mass $m > 0$ and charge $q \in \mathbb{R}$ is the Lorentzian manifold $(\mathbb{R} \times \mathbb{R}^n \setminus \{0\}, \mathfrak{g}_{m,q})$ where

$$\mathfrak{g}_{m,q} = -V_{m,q}(r)^2 dt^2 + V_{m,q}(r)^{-2} dr^2 + r^2 g_{\mathbb{S}^{n-1}},$$

where $g_{\mathbb{S}^{n-1}}$ denotes the standard metric on $\mathbb{S}^{n-1}$.

The n-dimensional Reissner–Nordström manifold is $\mathbb{R}^n \setminus \{0\}$, the spatial slice of this spacetime described by the metric

$$g_{m,q} = V_{m,q}(r)^{-2} dr^2 + r^2 g_{\mathbb{S}^{n-1}}.$$

We can rewrite the equations in isotropic coordinates as follows:

$$g_{m,q} = \left(1 + \frac{m+q}{2s^{n-2}}\right)^{\frac{2}{n-2}} \left(1 + \frac{m-q}{2s^{n-2}}\right)^{\frac{2}{n-2}} \delta =: \varphi_{m,q}^2 \delta,$$

$$V_{m,q}(s) = \frac{\left(1 - \frac{m^2-q^2}{4s^{2(n-2)}}\right)}{\left(1 + \frac{m+q}{2s^{n-2}}\right)\left(1 + \frac{m-q}{2s^{n-2}}\right)},$$

$$E_{m,q}(s) = \frac{n-2}{\mathfrak{C}_n} \frac{q\left(1 - \frac{m^2-q^2}{4s^{2(n-2)}}\right)}{s^{n-1}\varphi_{m,q}^{2n-3}(s)\left(\varphi_{m,q}(s) + s\varphi_{m,q}'(s)\right)} \partial_s,$$

where the coordinate transformation was performed by the the rule

$$r(s) = s\left(1 + \frac{m+q}{2s^{n-2}}\right)^{\frac{1}{n-2}} \left(1 + \frac{m-q}{2s^{n-2}}\right)^{\frac{1}{n-2}}.$$

Remark 3.1. A Reissner–Nordström manifold which is an electrostatic manifold can also be described using *electric potential* Ψ as in [Jah19, BCC24]. The connection between the two formulations is given by

$$E_{m,q} = \frac{n-2}{\mathfrak{C}_n}\frac{q}{r^{n-1}}V_{m,q}(r)\partial_r, \quad \Psi_{m,q} = \frac{1}{\mathfrak{C}_n}\frac{q}{r^{n-2}},$$

$$d\Psi_{m,q} = -\frac{n-2}{\mathfrak{C}_n}\frac{q}{r^{n-1}}dr, \quad \nabla^g\Psi = (d\Psi_{m,q})^\sharp = -\frac{n-2}{\mathfrak{C}_n}\frac{q}{r^{n-1}}V_{m,q}(r)^2\partial_r,$$

$$E_{m,q} = -\frac{\nabla^g\Psi_{m,q}}{V_{m,q}(r)}, \quad E^\flat_{m,q} = -\frac{(\nabla^g\Psi_{m,q})^\flat}{V_{m,q}(r)} = -\frac{d\Psi_{m,q}}{V_{m,q}(r)},$$

$$|E_{m,q}|^2 = \langle E_{m,q}, E_{m,q}\rangle = \langle (E^\flat_{m,q})^\sharp, (E^\flat_{m,q})^\sharp\rangle = \frac{|\nabla^g\Psi_{m,q}|^2}{V_{m,q}(r)^2} = \frac{|d\Psi_{m,q}|^2}{V_{m,q}(r)^2},$$

$$|E_{m,q}| = \sqrt{|E_{m,q}|^2} = \frac{n-2}{\mathfrak{C}_n}\frac{|q|}{r^{n-1}}.$$

Let $\psi = \dfrac{r}{n-2}$, then we have

$$|E_{m,q}|^2 = \left(\frac{n-2}{r}\right)^2\Psi^2_{m,q} = \psi^{-2}\Psi^2_{m,q} \quad \Longrightarrow \quad \Psi^2_{m,q} = \psi^2|E_{m,q}|^2,$$

$$|E_{m,q}| = \left(\frac{n-2}{r}\right)|\Psi_{m,q}| = \psi^{-1}|\Psi_{m,q}| \quad \Longrightarrow \quad |\Psi_{m,q}| = \psi|E_{m,q}|,$$

where $|\cdot| := |\cdot|_{g_{m,q}}$ for readability purposes.

Also, the equation for Reissner–Nordström electric potential $\Psi_{m,q}$ in isotropic coordinates is

$$\Psi_{m,q}(s) = \frac{q}{\mathfrak{C}_n s^{n-2}\left(1+\frac{m+q}{2s^{n-2}}\right)\left(1+\frac{m-q}{2s^{n-2}}\right)}.$$

Remark 3.2. Note that

$$\varphi_{m,q} = \left(\frac{(V_{m,q}+1)^2 - \mathfrak{C}_n^2\psi^2|E_{m,q}|^2_{g_{m,q}}}{4}\right)^{-\frac{1}{n-2}} = \left(\frac{(V_{m,q}+1)^2 - \mathfrak{C}_n^2\Psi^2_{m,q}}{4}\right)^{-\frac{1}{n-2}}. \tag{3.1}$$

3.2 Asymptotically flat systems

Another important concept for the uniqueness theorems is that of *asymptotically flat systems*. We begin by defining *asymptotically Reissner–Nordström systems*, followed by a definition of asymptotically flat systems. Although the former are asymptotically flat in the usual sense, in this paper, as in [BCC24], we call a system *asymptotically flat* if it satisfies asymptotic conditions weaker than those of the standard convention (see [BCC24, Remark 2.1] for further discussion of assumptions on asymptotic decay).

When proving uniqueness theorems, it will be more convenient for us to work with the electrostatic manifolds with boundary defined using electric potential Ψ. Substituting $VE = -\nabla^g\Psi$ into the system from Definition 2.3, we obtain the following definition of electrostatic manifolds with boundary:

Definition 3.1. The quadruple (M^n, g, V, Ψ) is an electrostatic manifold with boundary with $\Lambda = 0$ if it satisfies the following system of equations:

$$\begin{cases} \operatorname{Hess}_g V = V \operatorname{Ric}_g + \dfrac{2}{V} d\Psi \otimes d\Psi - \dfrac{2}{(n-1)} \dfrac{1}{V} |d\Psi|_g^2 \, g & \text{in } M, \qquad \text{(PEM 1)} \\ \Delta_g V = \dfrac{\mathfrak{C}_n^2}{V} |d\Psi|_g^2 & \text{in } M, \qquad \text{(PEM 2)} \\ 0 = \operatorname{div}_g \left(\dfrac{\nabla^g \Psi}{V} \right) & \text{in } M, \qquad \text{(PEM 3)} \\ \dfrac{\partial V}{\partial \nu} g - V B_g = 0 & \text{on } \partial M, \qquad \text{(PEM 4)} \end{cases}$$

where the equation $d(VE^\flat) = 0$ from (E3) is satisfied automatically due to $dd\Psi = 0$.

Also, the equations (NE1) and (NE2) transform into

$$\mathrm{R}_g = \frac{2|d\Psi|_g^2}{V^2} \qquad \text{in } M, \qquad \text{(NPEM 1)}$$

$$B_g = \frac{H_g}{n-1} g \qquad \text{on } \partial M. \qquad \text{(NPEM 2)}$$

For the sake of completeness, we proceed with the following definition from [CJM23, Definition 2.7].

Definition 3.2. A quadruple (M^n, g, V, E) with $n \geqslant 3$ is an *asymptotically Reissner–Nordström system of mass m and charge q*, if (M, g) is an *asymptotically Reissner–Nordström manifold of mass m and charge q*, i.e., there exists a compact set K in M, such that $M \setminus K = \sqcup_{k=1}^K M_k$, where M_k (called *end*) is diffeomorphic to $\mathbb{R}^n \setminus \overline{\mathbb{B}^n}$, $\Phi_k = (x^i) : M_k \to \mathbb{R}^n \setminus \overline{\mathbb{B}^n}$, and if, as $|x| \to \infty$, the following equations are satisfied

$$(\Phi_{k_*} g)_{ij} - (g_{m,q})_{ij} = O_2(|x|^{-(n-1)}),$$

$$\Phi_{k_*} V - V_{m,q} = O_2(|x|^{-(n-1)}),$$

$$\Phi_{k_*} \Psi - \Psi_{m,q} = O_2(|x|^{-(n-1)}),$$

where $|\cdot|$ denotes the Euclidean norm and $(g_{m,q})_{ij}, V_{m,q}, \Psi_{m,q}$ are all in isotropic coordinates.

As in [BCC24] we define 3-dimensional asymptotically flat electrostatic systems with weaker asymptotic decay.

Definition 3.3. An electrostatic manifold with compact connected boundary (M^3, g, V, Ψ) with nonvanishing V is asymptotically flat if there exists a compact set K in M with $\partial M \subset K$ such that $M \setminus K$ is diffeomorphic to $\mathbb{R}^3 \setminus \overline{\mathbb{B}^3}$, $\Phi_k = (x^i) : M_k \to \mathbb{R}^n \setminus \overline{\mathbb{B}^3}$ and, if, $|x|_\delta \to \infty$, the following equations are satisfied

$$g_{ij} = (\delta_{ij} + \eta_{ij}) \, dx^i \otimes dx^j, \quad \eta_{ij} = O_1(1),$$

$$V = 1 - \frac{\mu}{|x|} + O_2(|x|^{-1}), \quad \Psi = O_1(1),$$

where $\mu \in \mathbb{R}$ is a constant called the *mass*. And for the case when $\mu = 0$, the following equation is additionally satisfied

$$\Psi = \frac{\kappa}{|x|} + O_2(|x|^{-1}),$$

where $\kappa \in \mathbb{R}$ is a constant called the *charge*.

Remark 3.3. As in [CG21], we say a smooth function $f : \mathbb{R}^n \setminus \overline{\mathbb{B}^n} \to \mathbb{R}$ satisfies $f = O_l(|x|^p)$ for $l \in \mathbb{Z}_{\geqslant 0}, p \in \mathbb{R}$ as $|x| \to \infty$ if there exists a constant $C > 0$ and a centered open ball $\mathbb{B}^n_1 \supseteq \mathbb{B}^n$ such that

$$|D^J f| \leqslant C|x|^{p-|J|} \quad \text{in } \mathbb{R}^n \setminus \overline{\mathbb{B}^n_1}$$

for every multi-index J such that $|J| \leqslant l$ $\left(\text{here } D^J f = \dfrac{\partial^{|J|} f}{\partial x_1^{j_1} ... \partial x_n^{j_n}}\right)$.

Remark 3.4. As in [CJM23, Remark 2.8] and [BCC24, Remark 2.2] we note that asymptotically flat (and, hence, asymptotically Reissner–Nordström) systems (with boundary) are necessarily metrically complete and geodesically complete (up to the boundary) with at most finitely many boundary components that are necessarily all closed.

3.3 Quasilocal photon spheres

We proceed with the definition of quasilocal photon spheres as in [Jah19] and [CJM23] (only without restriction of the global existence of the electric potential Ψ), and then in Lemma 3.6 we prove a key result relating the boundary of the electrostatic manifold with boundary to photon spheres, which will be instrumental in the proofs of the uniqueness theorems.

Definition 3.4. Let (M^n, g, V, E) be an electrostatic manifold (with boundary). Then, a connected component S of a hypersurface in M is called a *quasilocal photon sphere* if it is totally umbilical, $H_S = const. < 0$, $V = const.$ on S, $|\mathrm{E}| = const.$ on S, and the equations

$$\mathrm{R}_S = \frac{n}{n-1} H_S^2 + 2|\mathrm{E}|^2, \tag{Q1}$$

$$\frac{\partial V}{\partial \nu} = \frac{H_S}{n-1} V \tag{Q2}$$

are satisfied on S.

Remark 3.5. In [Jah19, Definition 15] and [Rau21] quasilocal photon spheres are defined such that H_S is a positive constant. The reason is that we consider the outward pointing unit normal ν, while in [Jah19, Definition 15] and [Rau21] ν points toward the asymptotic end.

Remark 3.6. Note that photon spheres are particular cases of timelike, totally umbilical hypersurfaces in an electrostatic spacetime with $dV \neq 0$, $V = const.$, $|E| = const.$ and E being a normal vector field called *non-degenerate equipotential photon surfaces* (see [CJM23, Lemma 5.9] and [BCC24, Section 4.3.]). Also, due to (NE1) the obtained static electro-vacuum spacetime $(\mathbb{R} \times M^n, \mathfrak{g})$ from Section 2.1 satisfies the null energy condition (also see [CG16, CG17, GM17]).

Definition 3.5. A quasilocal photon sphere S in an electostatic manifold (with boundary) is called *subextremal* if

$$H_S^2 > \frac{\mathfrak{C}_n^2}{2}\,\mathrm{R}_S\,. \tag{3.3}$$

If instead of "$>$" in (3.3) there is "$=$" or "$<$", then S is called *extremal* or *superextremal* respectively.

Lemma 3.6. Let (M^n, g, V, E) be an electrostatic manifold with compact boundary with $\Lambda = 0$ and let S be a connected component of ∂M such that $V\big|_S = const. > 0$, $H_S < 0$, $E \in \Gamma\,(\mathrm{N}S)$ – normal vector field on S, and $|\mathrm{E}| = const.$ on S. Then S is a quasilocal photon sphere.

Proof.

1. From the condition $V\big|_S \neq 0$ and (NE2) it follows that S is umbilical and $\left.\dfrac{\partial V}{\partial \nu}\right|_S = const. < 0$.

2. (TE2) holds since (M^n, g, V, E) is an electrostatic manifold with boundary and, hence, (Q2) is satisfied.

3. Since (M^n, g, V, E) is an electrostatic manifold with boundary, S is a connected component of ∂M and $E \in \Gamma\,(\mathrm{N}S)$, then by Lemma 2.8 we have $H_S = const.$

4. Now we shall prove the equality (Q1).

   4.1. Using the contracted Gauss equation and umbilicity of S we obtain

$$\begin{aligned} R_g - 2\,\mathrm{Ric}_g(\nu,\nu) &= \mathrm{R}_S + |B_S|^2 - H_S^2 = \\ &= \mathrm{R}_S + \left|\frac{H_S}{n-1} g\right|^2 - H_S^2 = \\ &= \mathrm{R}_S + (n-1)\frac{H_S^2}{(n-1)^2} - H_S^2 = \\ &= \mathrm{R}_S - \frac{n-2}{n-1} H_S^2. \end{aligned} \tag{3.4}$$

   4.2. Also the formula for the Laplacian of a hypersurface implies

$$\Delta_g V = \Delta_S V + \mathrm{Hess}_g\, V(\nu,\nu) + H_S \frac{\partial V}{\partial \nu}. \tag{3.5}$$

   4.3. Then the condition $V\big|_S = const. > 0$, (3.5) and (E2) suggest

$$\begin{aligned} \mathrm{Hess}_g\, V(\nu,\nu) &= \Delta_g V - \Delta_S V - H_S \frac{\partial V}{\partial \nu} = \\ &= V\mathfrak{C}_n^2 |\mathrm{E}|_g^2 - \frac{2V}{n-1}\Lambda - 0 - H_S \frac{\partial V}{\partial \nu} = \\ &= V\mathfrak{C}_n^2 |\mathrm{E}|_g^2 - \frac{2V}{n-1}\Lambda - \frac{H_S^2}{n-1} V. \end{aligned} \tag{3.6}$$

4.4 Let us evaluate (E1) in the normal direction ν to S:

$$\begin{aligned}\operatorname{Hess}_g V(\nu,\nu) &= V\operatorname{Ric}_g(\nu,\nu) - \frac{2V}{n-1}\Lambda g(\nu,\nu) - \\ &+ 2V\left(\mathrm{E}^\flat\otimes\mathrm{E}^\flat\right)(\nu,\nu) - \frac{2}{n-1}V|\mathrm{E}|_g^2\, g(\nu,\nu) = \\ &= V\operatorname{Ric}_g(\nu,\nu) - \frac{2V}{n-1}\Lambda + 2V|\mathrm{E}|_g^2 - \frac{2}{n-1}V|\mathrm{E}|_g^2 = \\ &= V\operatorname{Ric}_g(\nu,\nu) - \frac{2V}{n-1}\Lambda + \frac{2(n-2)}{n-1}V|\mathrm{E}|_g^2 = \\ &= V\operatorname{Ric}_g(\nu,\nu) - \frac{2V}{n-1}\Lambda + \mathfrak{C}_n^2 V|\mathrm{E}|_g^2,\end{aligned} \tag{3.7}$$

where in the second line we used the condition that $E\in\Gamma(\mathrm{N}S)$.

4.5 Combining (3.7), (3.6) and the fact that $V\big|_S = const. > 0$ we obtain on S

$$\operatorname{Ric}_g(\nu,\nu) = -\frac{H_S^2}{n-1}. \tag{3.8}$$

4.6 Finally (3.8), (3.4) and (NE1) with $\Lambda = 0$ imply on S

$$\begin{aligned}\mathrm{R}_S &= \frac{n-2}{n-1}H_S^2 + \frac{2}{n-1}H_S^2 + 2|\mathrm{E}|_g^2 + 2\Lambda = \\ &= \frac{n}{n-1}H_S^2 + 2|\mathrm{E}|_g^2 + 2\Lambda,\end{aligned} \tag{3.9}$$

which is exactly the equation (Q1) with $\Lambda = 0$.

□

4 Proofs of the uniqueness theorems

Analogously to [Med24, Example 6], we are to consider n-dimensional Reissner–Nordström space RN^n, where RN^n with $m > |q|$ is called *subextremal*, RN^n with $m = q$ is called *extremal* and RN^n with $m < |q|$ is called *super-extremal*.

The space RN^n with $m \geqslant |q|$ has a a unique photon sphere at r_{ps}^+ (see Section 5) which splits it into a compact part and a non-compact part containing the end. We denote the latter part by RN^n_- and note that it is an electrostatic manifold with boundary with nonvanishing static potential.

We can also get another example of a non-compact electrostatic manifold with boundary by doubling the space RN^n with $m \geqslant |q|$ and considering the part of its domain bounded by the image of the photon sphere and containing RN^n. We denote this manifold by RN^n_+.

Reissner–Nordström space with $|q| > m$ is called *super-extremal* and has two photon spheres when $|q| > m > \frac{2\sqrt{n-1}}{n}|q|$, a unique photon sphere when $m = \frac{2\sqrt{n-1}}{n}|q|$ and no photon spheres otherwise. By analogy, for the cases with either two photon spheres or

a unique photon sphere $RN^n_- = ([r^+_{ps}, \infty) \times \mathbb{S}^{n-1}, g_{m,q}, V_{m,q}(r), E_{m,q}(r))$ (see Appendix 5). Also, we denote by r_0 some nonnegative constant chosen such that $[r_0, \infty)$ is the maximal interval on which the static potential $V_{m,q}$ is well defined (i.e. $V_{m,q}$ is well defined for all $r > r_0$)(see [BCC24, Section 2]). Then, for the super-extremal Reissner–Nordström manifold with no photon spheres $RN^n_- = ([r_+, \infty) \times \mathbb{S}^{n-1}, g_{m,q}, V_{m,q}(r), E_{m,q}(r))$, where r_+ is some constant such that $r_+ > r_0$. Again, from [BCC24, Section 2] we know that there exists a spherically symmetric equipotential photon surface going through $\{r = r_+\}$ which corresponds to the boundary of the electrostatic manifold.

Now we are ready to prove the uniqueness theorems.

Proof of Theorem 1.1.

I.

1| If $V \neq 0$ on $M \cup \partial M$, then since V is an asymptotically Reissner–Nordström static potential we have $V > 0$ on M and, hence, $V = const. > 0$ on ∂M.

2| Since (M, g) is asymptotically Reissner–Nordström, (M, g, V, E) is an electrostatic manifold with boundary ∂M consisting of subextremal connected components, $|\mathrm{E}| = const.$ on ∂M and $E \in \Gamma(\mathrm{N}\partial M)$, the first statement follows from Lemma 3.6 and the Reissner–Nordström photon sphere uniqueness theorem [Jah19, Theorem 3].

II.

1| Now let us consider (M^3, g, V, Ψ) with $\Sigma = V^{-1}(0) \subset \operatorname{Int} M$ compact, and separating the boundary and the end.

2| Cut M along Σ.

3| Let N be the non-compact part of $M \setminus \Sigma$. On N we have $V \geqslant 0$ with $\partial N = \Sigma$ being totally geodesic by Lemma 2.8 and $\dfrac{\partial V}{\partial \nu} = const. \neq 0$. Then $\partial N = \Sigma$ is a non-degenerate static horizon so by [Jah19, Corollary 4] N is isometric to RN^3_-, ∂N is connected and isometric to $\{m + \sqrt{m^2 - q^2}\} \times \mathbb{S}^2$.

4| Let Ω be the compact part of $M \setminus \Sigma$, then $\partial\Omega$ has two connected components:

i) $\Sigma = V^{-1}(0)$ totally geodesic round sphere $\{m + \sqrt{m^2 - q^2}\} \times \mathbb{S}^2$,

ii) S – mean convex round sphere:

- (Q1) holds so $\mathrm{R}_S > 0$ and, since for a surface the scalar curvature is equal to doubled Gaussian curvature, then by the Gauss–Bonnet theorem $\chi(S) > 0$ and, due to S being closed and orientable, we have $\chi(S) = 2 - 2\mathbf{g} > 0$ implying $\mathbf{g} = 0$ and, hence, S is a topological sphere;
- S is round mean convex since $H_S = const$ with $V\big|_S < 0$.

5| The rest of the proof generally follows the same arguments as those used in the end of the proof of [Jah19, Theorem 3]:

5.1| Attach RN^3_- to $\partial\Omega$ along S (after a possible rescaling of m and q) and denote this manifold by $\widetilde{M}$.

5.2 Remember that an electrostatic manifold with boundary (M, g, V, E) could be redefined as (M, g, V, Ψ) with $VE = -\nabla^g \Psi$ (see Definition 3.1). Then, define a Riemannian metric, static potential and electric potential $\widetilde{g}$, $\widetilde{V}$, and $\widetilde{\Psi}$ respectively as

$$\widetilde{g} = \begin{cases} g, & \text{on } \Omega, \\ g_{m,q}, & \text{on } RN^3_-, \end{cases}$$
$$\widetilde{V} = \begin{cases} V, & \text{on } \Omega, \\ -V_{m,q}, & \text{on } RN^3_-, \end{cases}$$
$$\widetilde{\Psi} = \begin{cases} \Psi, & \text{on } \Omega, \\ \Psi_{m,q}, & \text{on } RN^3_-, \end{cases}$$

where $g_{m,q}, V_{m,q}$ and $\Psi_{m,q}$ are the metric, the static potential potential and the electric potential resp. on RN^3_-. Notice that $\widetilde{g}, \widetilde{V}$ and $\widetilde{\Psi}$ are smooth away from S and $C^{1,1}$ across it.

5.3 Then $\left(\widetilde{M}, \widetilde{g}, \widetilde{V}, \widetilde{\Psi}\right)$ is a non-compact $C^{1,1}$-electrostatic manifold with boundary.

5.4 Doubling the manifold by reflection across Σ we obtain an asymptotically Reissner–Nordström $C^{1,1}$-manifold $\left(\widehat{M}, \widehat{g}, \widehat{V}, \widehat{\Psi}\right)$ which has two ends.

5.5 We then perform a conformal change of the metric $\widehat{g}$ as $\widehat{\widehat{g}} := \Theta^2 \widehat{g}$, where Θ denotes the conformal factor defined as follows

$$\Theta := \frac{(1+\widehat{V})^2 - \widehat{\Psi}^2}{4}.$$

5.6 Define on $\widehat{M}$ the smooth function $F_\pm := \widehat{V} - 1 \pm \widehat{\Psi}$ and set $\widehat{M}_+ = \{p \in \widehat{M} : \widehat{V}(p) \geqslant 0\}$, $\widehat{M}_- = \{p \in \widehat{M} : \widehat{V}(p) \leqslant 0\}$. From [KL18, Lemma 2] and [Jah19, Lemma 24 and Section 6.3.] we know that $F_\pm < 0$ on $\widehat{M}_+$. Then, $0 < (1+\widehat{V})^2 - \widehat{\Psi}^2 = 4\Theta$ on $\widehat{M}$ and, hence, $\Theta > 0$ on all of $\widehat{M}$. Consequently, all conditions required by [Jah19, Proposition 21 and Proposition 22] hold. Using [Jah19, Proposition 22], we compactify the non-reflected end of $\left(\widehat{M}, \widehat{\widehat{g}}\right)$ by adding a point at infinity and obtain a one-ended geodesically complete, scalar flat manifold $\left(\widehat{M}_\infty, \widehat{\widehat{g}}_\infty\right)$ which is C^∞ away from the gluing surfaces and the inserted point at infinity, and $C^{1,1}$ along them. Moreover, by [Jah19, Proposition 21] it has zero ADM-mass and zero charge, and by [KL18] its scalar curvature is given by

$$\frac{1}{8\widehat{V}\Theta^2}\left|2\widehat{V}\widehat{\Psi}\nabla\widehat{V} - (\widehat{V}^2 - 1 + \widehat{\Psi}^2)\nabla\widehat{\Psi}\right|^2 \geqslant 0.$$

5.7 Thus by the rigidity case of Bartnik's positive mass theorem [Bar86], $\left(\widehat{M}_\infty, \widehat{\widehat{g}}_\infty\right)$ is the Euclidean 3-space.

5.8 Hence, $\left(\widehat{M}, \widehat{g}\right)$ is C^∞ and conformally flat, part of which coincides with RN^3_-. Then it follows from [Jah19, Section 6.4.] and [KL18] that this is the doubling of the subextremal Reissner–Nordström 3-manifold. Thus, Ω is the reflection of $\left([r_h, r_{ps}] \times \mathbb{S}^2, g_{m,q}\right)$ across the boundary. Therefore, (M, g) is subextremal RN^3_+, $V = V_{m,q}$ and $E = E_{m,q}$.

□

Proof of Theorem 1.2.

1. If $V \neq 0$ on $M \cup \partial M$, then since V is an asymptotically flat static potential, without loss of generality we have $V > 0$ on M and, hence, $V = const. > 0$ on ∂M.

2. Since (M^3, g, V, E) is an asymptotically flat electrostatic system which is an electrostatic manifold with connected boundary ∂M, $|\mathrm{E}| = const.$ on ∂M, $E \in \Gamma(\mathrm{N}\partial M)$ and $V^2 \geqslant |1 - \Psi^2|$ on ∂M holds if $V^2 > (1 - |\Psi|)^2$ on ∂M unless $V = 1$ and $\Psi = 0$ on ∂M, the first statement follows from Lemma 3.6 and the electrostatic equipotential photon surface uniqueness theorem [BCC24, Theorem 3.2.].

□

Appendix

A piece of the Reissner–Nordström manifold with positive mass is an example of an electrostatic manifold with boundary with cosmological constant $\Lambda = 0$. Using (E4) and $V_{m,q}$ from Section 3.1 we shall explicitly derive the photon sphere equation.

Fix two real parameters $m > 0$ and q (the mass and the charge respectively). Then the Reissner–Nordström space is the Riemannian manifold $\big(M = [r_0, \infty) \times \mathbb{S}^{n-1}, g_{m,q}\big)$, where $r_0 \geqslant 0$ such that $V_{m,q}$ is well-defined for all $r > r_0$.

We shall solve (E4) for Reissner–Nordström manifold:

$$\frac{\partial V_{m,q}}{\partial \nu_{m,q}} g - V_{m,q} B_g = 0 \quad \text{on } \Sigma_r.$$

Firstly, let $\nu_{m,q}$ denote the *unit normal* to some hypersurface in the Reissner–Nordstrom manifold. We can write it as $\nu_{m,q} = f(r)\partial_r$ with some arbitrary smooth function f. Then we have $1 = g(\nu_{m,q}, \nu_{m,q}) = f(r)^2 g_{m,q}(\partial_r, \partial_r) = f(r)^2 (g_{m,q})_{rr} = f(r)^2 V_{m,q}(r)^{-2}$. Therefore, $f(r) = V_{m,q}(r)$ and, hence, $\nu_{m,q} = V_{m,q}(r)\partial_r$. Then, the derivative of the static potential $V_{m,q}$ in the normal direction is given by

$$\frac{\partial V_{m,q}}{\partial \nu_{m,q}} = V_{m,q} \partial_r \left(V_{m,q}\right) = \frac{1}{2}\frac{d}{dr}\left(1 - \frac{2m}{r^{n-2}} + \frac{q^2}{r^{2(n-2)}}\right) = (n-2)\left(\frac{m}{r^{n-1}} - \frac{q^2}{r^{2n-3}}\right). \tag{5.1}$$

Notice that

$$\begin{aligned}\sqrt{\det g_{m,q}} &= \sqrt{(g_{m,q})_{rr} \det(r^2 g_{\mathbb{S}^{n-1}})} = \\ &= \sqrt{V_{m,q}(r)^{-2} r^{2(n-1)} \det g_{\mathbb{S}^{n-1}}} = \\ &= V_{m,q}(r)^{-1} r^{n-1} \sqrt{\det g_{\mathbb{S}^{n-1}}}.\end{aligned}$$

Then, let us calculate the mean curvature of Σ_r:

$$\begin{aligned}H_{\Sigma_r} &= \operatorname{div}_{\Sigma_r} \nu_{m,q} = \operatorname{div}_{\Sigma_r} \left(V_{m,q}(r)\partial_r\right) = \\ &= \frac{1}{\sqrt{\det g_{m,q}}} \partial_r \left(V_{m,q}(r)\sqrt{\det g_{m,q}}\right) = \\ &= \left(V_{m,q}(r)^{-1} r^{n-1} \sqrt{\det g_{\mathbb{S}^{n-1}}}\right)^{-1} \partial_r \left(r^{n-1}\sqrt{\det g_{\mathbb{S}^{n-1}}}\right) = \\ &= \frac{V_{m,q}(r)}{r^{n-1}\sqrt{\det g_{\mathbb{S}^{n-1}}}} \sqrt{\det g_{\mathbb{S}^{n-1}}}(n-1) r^{n-2} = \\ &= \frac{(n-1)V_{m,q}(r)}{r}.\end{aligned}$$

Consequently, using (5.1) we obtain

$$\begin{aligned}
&\frac{\partial V_{m,q}}{\partial \nu_{m,q}}(n-1) - V_{m,q} H_{\Sigma_r} = 0 \quad \Longrightarrow \\
&\Longrightarrow \quad (n-2)\left(\frac{m}{r^{n-1}} - \frac{q^2}{r^{2n-3}}\right)(n-1) - \frac{(n-1)V_{m,q}^2}{r} = 0 \quad \Longrightarrow \\
&\Longrightarrow \quad \frac{(n-2)m}{r^{n-1}} - \frac{(n-2)q^2}{r^{2n-3}} - \frac{1}{r}\left(1 - \frac{2m}{r^{n-2}} + \frac{q^2}{r^{2(n-2)}}\right) = 0 \quad \Longrightarrow \\
&\Longrightarrow \quad \frac{(n-2)m}{r^{n-1}} - \frac{(n-2)q^2}{r^{2n-3}} - \frac{1}{r} + \frac{2m}{r^{n-1}} - \frac{q^2}{r^{2n-3}} = 0 \quad \Longrightarrow \\
&\Longrightarrow \quad -\frac{1}{r} + \frac{nm}{r^{n-1}} - \frac{(n-1)q^2}{r^{2n-3}} = 0 \quad \Longrightarrow \\
&\Longrightarrow \quad r^{2(n-2)} - (nm)r^{n-2} + (n-1)q^2 = 0.
\end{aligned}$$

After substitution $u = r^{n-2}$ the equation becomes

$$u^2 - (nm)u + (n-1)u = 0.$$

The solution is

$$u_{ps} = \frac{1}{2}nm \pm \frac{1}{2}\sqrt{n^2m^2 - 4(n-1)q^2}.$$

Substituting $u = r^{n-2}$ back into the equation, we obtain two solutions

$$r_{ps}^{+} = \left(\frac{1}{2}nm + \frac{1}{2}\sqrt{n^2m^2 - 4(n-1)q^2}\right)^{\frac{1}{n-2}}, \tag{5.2}$$

$$r_{ps}^{-} = \left(\frac{1}{2}nm - \frac{1}{2}\sqrt{n^2m^2 - 4(n-1)q^2}\right)^{\frac{1}{n-2}}. \tag{5.3}$$

From [CJM23, Corollary 3.2] we know that for Reissner–Nordström manifold if $m > |q|$ or $m = |q| \neq 0$, then it has a unique photon sphere which lies at r_{ps}^{+}. This sphere is known in the literature as (a time-like slice of) the *photon sphere* (see the definitions in [CVE01, Per05]). If $|q| > m > \frac{2\sqrt{n-1}}{n}|q|$, then it has two photon spheres at r_{ps}^{+} and r_{ps}^{-}, and if $m = \frac{2\sqrt{n-1}}{n}|q|$, then it has a unique photon sphere at $r_{ps} = r_{ps}^{+} = r_{ps}^{-}$.

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